

ADBMS Mid Term Examination Revision

Ques.1: What is Data and Database? Explain types of databases.

➔ **Data:** Representation of facts, figures, statistics having no particular meaning. It can be in the form of numbers, characters, symbols or even pictures.

Information: Processed data which has a perfect meaning.

Database: Database is a collection of related data, i.e. known facts that can be recorded and have implicit meaning. It represents some aspect of the real world. It is a logically coherent collection of data with some inherent meaning which is designed built & populated with data for a specific purpose.

Types of databases:

- Traditional Database
- Multimedia Database
- Geographic Information System (GIS)
- Data warehouses and on-line analytical processing (OLAP)
- Real time and active Database

Ques.2: What is DBMS? Explain the characteristic, advantages and use case of database approach.

➔ DBMS or Database Management System is a collection of programs that enables users to create and maintain the database. It is a general purpose software system that facilitates the process of defining, constructing, and manipulating a database for various applications.

- **Defining:** To specify the data types, structures & constraints of data to be stored in the database.
- **Constructing:** To store the data itself on some storage medium that is controlled by the DBMS.
- **Manipulating:** Querying the database to retrieve specific data updating the database to reflect changes in number and generating reports of data.

Characteristics of Database

- Data Integrity
- Scalability
- Concurrency
- Data Abstraction
- Normalization
- Self-Describing
- Multi View
- Data Sharing

Advantages over File Approach

- Redundancy Control
- Better Security
- Backup & recovery
- Easy access and management
- Independence

Limitations of Database Approach

- High Cost
- Database Failure
- Frequent Update Cycle
- Complexity
- Performance Issues

Ques.3: Classification of DBMS based on Data Models.

➔ A data model is a conceptual representation of how data is being structured, organized and manipulated within the database. It defines the relationship between different data elements and helps in understanding how data can be stored, retrieved and interacted with. Following are the types of data model:

- **Hierarchical Model:** Data is organized in a tree like structure, with a single parent for each record. Simple but inflexible.
- **Network Model:** Similar to hierarchical model but allows more complex relationships with multiple parent records. Data can be accessed in various ways.
- **Relational Model:** Data is organized in a table with rows(records) and columns(attributes). Most widely used model today, used in SQL for data manipulation.
- **Object Oriented Model:** Data is represented as objects similar to object-oriented programming. Supports complex data types and relationships.
- **ER Model:** Entity Relationship Model, used primarily in database design. It visually depicts the structure of the database.
- **Document Model:** Used in NoSQL databases, data is stored in documents(often JSON or XML). This model is flexible and allows for unstructured data.

Ques.4: Explain DBMS Architecture?

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Ques.5: What is three schema architecture?

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Ques.6: Explain Data Independence and Data Abstraction.

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Ques.7: What is ER Model with example?

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Ques.8: What are different keys in a database?

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Ques.9: What are some DBMS users?



Ques.10: What are DBMS languages?



Ques.11: DBMS Interfaces. Explain.



Ques.12: What is Normalization? Explain different types of normalization.



Ques.13: What is Join in DBMS? Explain its types.

➔ In a Database Management System (DBMS), a join is an operation that combines rows from two or more tables based on a related column between them. Joins allow you to retrieve related data from multiple tables in a single query.

Types of Joins

1. Inner Join:

- Returns only the rows that have matching values in both tables.
- Example: If you join a "Customers" table and an "Orders" table on a "CustomerID" column, you'll only get the customers who have placed orders.

2. Outer Joins:

- These return all rows from one table and the matching rows from the other table. If there is no match, NULL values are returned for the non-matching side.
- Types of Outer Joins:
 - Left Outer Join (or Left Join): Returns all rows from the left table and matched rows from the right table. Unmatched rows from the right table will show NULL.
 - Right Outer Join (or Right Join): Returns all rows from the right table and matched rows from the left table. Unmatched rows from the left table will show NULL.
 - Full Outer Join: Returns all rows when there is a match in either table. Rows from both tables will show NULL for non-matching entries.

3. Cross Join:

- Produces a Cartesian product of the two tables, returning all possible combinations of rows. This can result in a very large result set if both tables have many rows.

4. Self-Join:

- A join where a table is joined with itself. This is useful for querying hierarchical data or comparing rows within the same table.

5. Natural Join:

- Automatically joins tables based on all columns with the same name and data type in both tables. It eliminates duplicate columns in the result.